Oral

Decoupled Diffusion Solver for Inverse Problems on Function Spaces

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Introduction

Motivation

- Inverse PDE problems aim to reconstruct unknown coefficient field $a(x)$ from sparse, partial observation of solution field $u(x)$. Yet, prior generative methods rely on joint-embedding of the coefficient $\boldsymbol{\mathcal{A}}$ and solution space $\boldsymbol{\mathcal{U}}$.

Key Contributions

- We build DDIS, decoupling prior from physics: a diffusion learns prior $p(a)$, and a neural operator models L: $a \mapsto u$;
- We prove two failure modes of joint-embedding: **guidance attenuation** and **sparse-guidance collapse**;
- DDIS achieves SOTA accuracy and data efficiency, leading by **11%+ L2**, **54%+ spectral error**, and **40%+** at **1% paired data**.

Background

Bayesian inverse PDE problems

- To infer coefficient field $a(x)$ from sparse observations $u_{obs} = L(a) + \epsilon$, i.e., posterior $p(a \mid u_{obs}) \propto p(a)$.

Diffusion-based posterior sampling

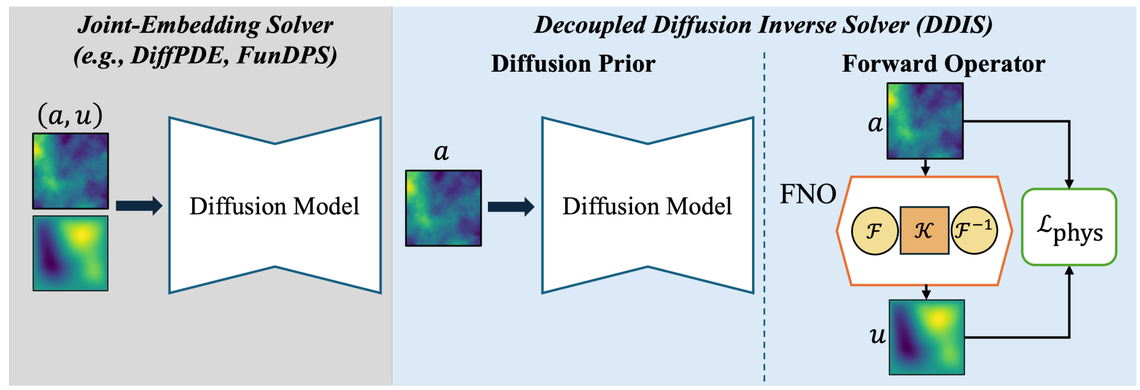
- Train the diffusion on the prior, while during inference use the observation to *guide* the sampling toward posterior.
- The canonical DPS method suffers over-smoothing due to Jensen-gap approximation against intractable formulation.

Previous Methods

- Directly learn joint $p(a, u)$, representing physics mapping L: $a \mapsto u$ implicitly and relies heavily on paired-(a, u) data.

Proposed Method

Learning framework (Joint-embedding solver vs. our DDIS)



Comparison Summary

	DDIS	Joint-Embedding Models
Training Stage		
Architecture	$G_\theta : \mathcal{Z} \to \mathcal{A}$ and $L_\phi : \mathcal{A} \to \mathcal{U}$	$H_\psi : \mathcal{Z} \to \mathcal{A} \times \mathcal{U}$
Objective	model $p(a)$ and learn $L(\cdot)$	model $p(a, u)$
Data Utilization	prior: $n_u + n_p$; operator: n_p	n_p only
Complexity‡	prior: d_P; operator: d_L	$d_J \geq \max(d_P, d_L)$
Generalization Bound‡	$\tilde{\mathcal{O}}\left(\sqrt{\frac{d_L}{n_p}} + \sqrt{\frac{d_P}{n_p + n_u}}\right)$	$\tilde{\mathcal{O}}\left(\sqrt{\frac{d_J}{n_p}}\right)$
Physics Repr.	explicit surrogate $L_\phi(a)$	implicit in joint model $p(a, u)$♭
Posterior Sampling Stage		
Target	$p(a \mid u_{\text{obs}})$	$p(a \mid u_{\text{obs}})$ via marginalizing out u
Sampler	DAPS (Zhang et al., 2025)	typically DPS (Chung et al., 2022)
Likelihood	$\log p(u_{\text{obs}} \mid \widehat{a}_{0\mid u})$	$\log p(u_{\text{obs}} \mid \mathbb{E}[a_0 \mid a_{t+1}])$
Bias	asymptotically unbiased†	Jensen-gap bias
Spectral Feature	preserves high-frequency details	degraded at high frequencies
Inference Cost	low, diffusion forward + FNO backward	high, diffusion forward and backward

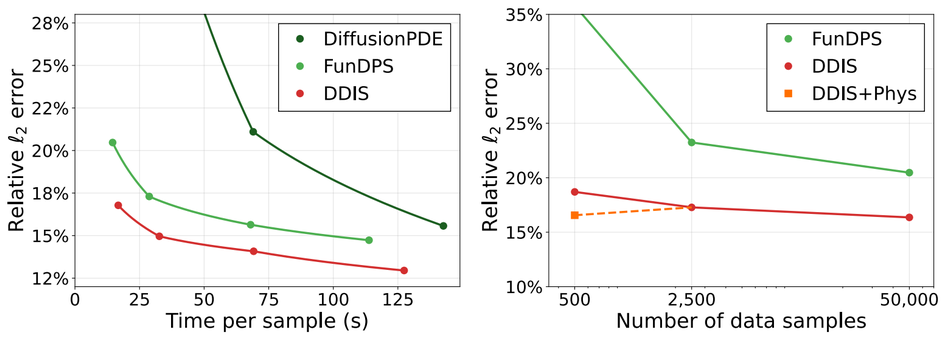
Results

Standard supervision results under 3% sparse observations

Method	Poisson ℓ_2	Poisson E_s	Helmholtz ℓ_2	Helmholtz E_s	N-S ℓ_2	N-S E_s	Time (s)
DiffusionPDE	74.68	.566	46.10	.315	32.78	2.099	17.75
FunDPS	19.96	.192	17.16	.140	8.99	.382	14.58
DDIS	**15.78**	**.074**	**15.08**	**.044**	**8.93**	**.165**	16.75
DiffusionPDE	35.53	.281	23.86	.164	11.01	.221	35.63
FunDPS	17.14	.135	16.05	.157	8.23	.423	28.75
DDIS	**14.36**	**.086**	**14.24**	**.055**	**8.02**	**.178**	32.67
DiffusionPDE	16.65	**.085**	17.73	**.096**	9.24	.219	142.62
FunDPS	14.73	.136	14.14	.198	7.98	.479	113.83
DDIS	**12.32**	.087	**12.20**	.097	**7.81**	**.188**	127.42
ECI-sampling*	94.63	/	92.83	/	42.36	/	0.20
ECI-sampling*	93.47	/	93.23	/	41.68	/	0.38
OFM*	71.87	/	49.60	/	37.57	/	470.44
OFM*	47.04	/	42.07	/	20.98	/	4366.34

- DDIS lies on the **accuracy-runtime Pareto frontier**;
- It attains lower spectral error, indicating less over-smoothing.

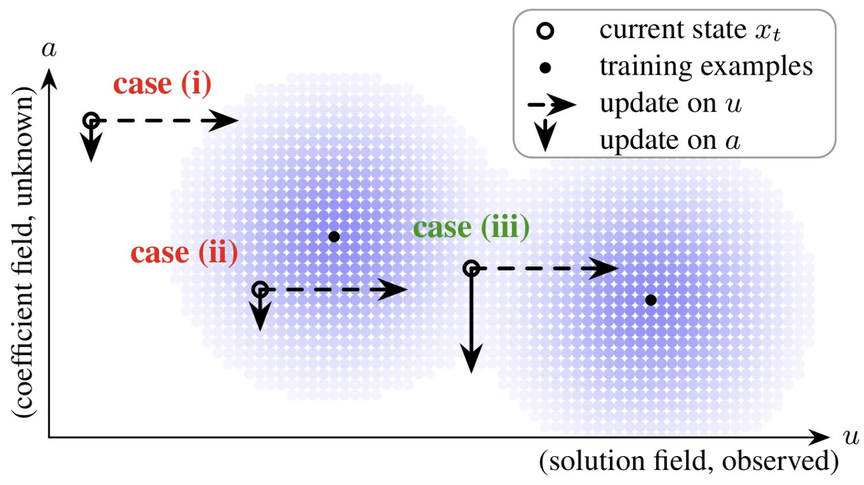
Scarce paired-data supervision (100%, 5%, and 1% data)

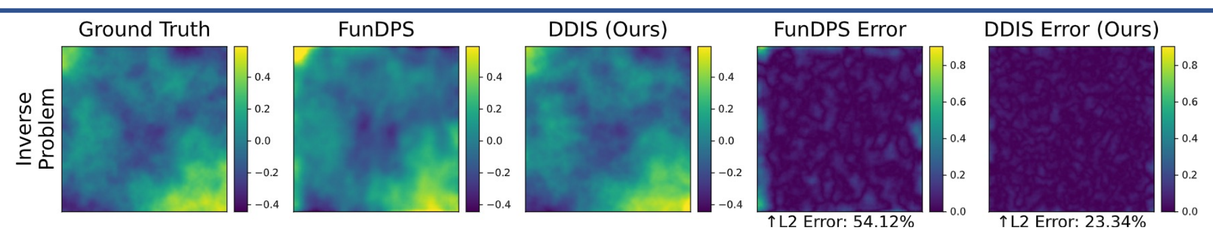


Data Scarcity	Poisson DDIS	Poisson Fun.‡	Helmholtz DDIS	Helmholtz Fun.‡	N-S DDIS	N-S Fun.‡	T (s)
100%	**16.36**	20.47	**15.19**	17.16	**8.22**	8.48	16
5%	**17.28**	23.24	**15.69**	20.97	**9.21**	11.56	
1%	**18.70**	35.81	**16.40**	41.69	**12.05**	13.65	
1%+Phys	**16.56**	35.81	**16.05**	41.69	/ †	/ †	

- Joint-embedding methods degrade as paired data becomes scarce;
- The failure is due to **guidance attenuation** shown below ↓;
- DDIS avoids this failure mode by decoupling.

Theory





❖ Guidance by joint-embedding vanishes under scarce paired data. ↑Overall empirical results.

← Blue blobs: regions supported by the learned joint model around training examples (black dots), and the circle marker: diffusion state x_t. We consider 3 cases: when x_t is far from all blobs (i), near a single training sample (ii), or close to multiple training samples (iii).

← In all cases, the update on u (dashed →) remains valid; yet, the update on a (solid →) vanishes in (i) and (ii), and is nonzero only in (iii). In high dimensions, case (iii) is rare, rendering the guidance ineffective.

Conclusion and Outlook

Thesis-Antithesis-Synthesis: Decoupling in classical inverse solver rely on hand-crafted or numerical guidance that is brittle under complex forward maps. Joint-embeddings remove the hand-crafting but entangle prior and physics, leading to guidance attenuation. DDIS restores decoupling with a physics-native surrogate.

Explicit physics beats learned correlation: NO surrogate replaces joint-embedding, avoiding guidance attenuation and improving data efficiency.

Multi-physics: DDIS extends naturally to multi-physics settings, by one shared coefficient prior and multiple surrogates representing each subsystem.

References

[1] (FunDPS). Yao, Jiachen, et al. "Guided Diffusion Sampling on Function Spaces with Applications to PDEs." The Thirty-ninth Annual Conference on Neural Information Processing Systems.
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[3] (Neural Operator). Kovachki, Nikola, et al. "Neural operator: Learning maps between function spaces with applications to pdes." Journal of Machine Learning Research 24.89 (2023): 1-97.
[4] (Score-matching). Lim, Jae Hyun, et al. "Score-based diffusion models in function space." JMLR 26.158 (2025): 1-62.

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